The second distinct volume of entirely ideal right-angled hyperbolic four-polytopes

Coxeter Polytopes Frontier Project

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Abstract

Among finite-volume entirely ideal right-angled hyperbolic four-polytopes, the smallest volume is $4\pi^2/3$, attained uniquely by the ideal 24-cell, by Kolpakov. We prove that the second distinct volume is $8\pi^2/3$. The new lower bound uses facet defects, the geometry of opposite faces of a cusp box, and the published classification of the nine-vertex ideal right-angled three-polyhedron. Reflection of the 24-cell across one facet attains the bound. An exact rational certificate checks this double and its complete face and cusp data. The universal geometric argument is a written proof, not a finite four-polytope census. This manuscript translates the internally checked argument; external refereeing and an end-to-end formalization are not claimed. Classification of all polytopes attaining the second value is not a theorem of this manuscript.

1 Statement and inputs

Throughout, a polytope is finite-sided, convex, four-dimensional, and of finite hyperbolic volume. "Entirely ideal" means that every vertex is on the sphere at infinity. "Right-angled" means that every dihedral angle is $\pi/2$. Let C denote the regular ideal right-angled 24-cell.

Theorem 1.1. *If P is an entirely ideal right-angled hyperbolic four-polytope and P is not isometric to C, then*

$$\mathrm{Vol}(P) \geq \frac{8\pi^2}{3}.$$

Equality is attained by the union of C and its reflection across an octahedral facet. Consequently the second distinct volume is $8\pi^2/3$.

The distinction between a second volume *value* and the classification of its equality cases is essential. Theorem 1.1 does not assert uniqueness at $8\pi^2/3$.

We use the following published inputs, with the indicated precise scope. Kolpakov's Lemmas 1–2 give the four-dimensional incidence and volume formulas, and Theorem 3 identifies C as the unique minimum [1]. Proposition 5 states that an ideal right-angled three-polyhedron with a k-gonal face has at least $2k$ vertices, with equality exactly for the k-antiprism. Section 4.3, Case D, and Figure 6 give the complete admissible small octahedrite list and, in particular, the unique nine-vertex case. We use only that nine-vertex classification, not a classification of arbitrary four-polytopes. The standard right-angled rigidity of these three-dimensional polyhedra identifies the classified combinatorial types with their geometric models, as in Kolpakov's use of Andreev's theorem.

The new opposite-cusp injection, cusp-shape calculation, minimum-defect argument and their combination are internal arguments below. They are not treated as published assumptions. The exact construction certificate has a separate scope from the universal proof.

2 Discrete volume and facet defect

Write v, e, r, f for the numbers of vertices, edges, ridges and facets of P. An ideal vertex link is a Euclidean rectangular three-box. Counting its vertices and edges and using Euler's identity gives

$$e = 4v, \qquad r = 3v + f, \qquad \sum_{R \in \Omega_2(P)} k(R) = 12v,$$

where $k(R)$ is the number of vertices of a ridge R. Define

$$W(P) = \sum_R (k(R) - 3).$$

It follows that

$$W(P) = 3(v - f) \in 3\mathbb{Z}_{\geq 0}, \qquad \mathrm{Vol}(P) = \frac{\pi^2}{3}(v - f + 4) = \frac{4\pi^2}{3} + \frac{\pi^2}{9} W(P). \tag{1}$$

The latter identity is Kolpakov's volume formula; equivalently, Gauss–Bonnet applied to the reflection orbifold uses $\chi = 1 - f/2 + r/4 - e/8 = (v - f + 4)/4$.

Every facet F is an ideal right-angled three-polyhedron. Its ideal vertices have valency four, so, writing v_F for their number, it has $2v_F$ edges and $v_F + 2$ polygonal faces. Its defect is

$$d(F) := v_F - 6 = \sum_{R \subset F} (k(R) - 3), \qquad \sum_F d(F) = 2W(P). \tag{2}$$

Defect zero characterizes the regular ideal octahedron. A positive defect facet has a nontriangular face of size $k \geq 4$; the antiprism bound $v_F \geq 2k$ implies $d(F) \geq 2$. In particular defect one does not occur.

3 Opposite cusp facets

For a facet F, let $q(F)$ be its number of nontriangular faces and $s(F)$ the number of its cusps whose intrinsic horospherical rectangle is nonsquare. The latter condition does not depend on horosphere scale.

Lemma 3.1 (Opposite-cusp injection)**.** *Suppose the smallest positive facet defect of P is m, and F has positive defect. There are $q(F) + s(F)$ pairwise distinct positive-defect facets other than F: its neighbors across nontriangular ridges, and the facets opposite F at its nonsquare cusps. Therefore*

$$2W(P) \geq d(F) + m\big(q(F) + s(F)\big). \tag{3}$$

In particular, when $d(F) = m$, $2W(P) \geq m(1 + q(F) + s(F))$.

Proof. A neighbor across a nontriangle contains that same nontriangle and is not octahedral. Distinct ridges of F have distinct neighboring facets, since the intersection of two facets is a face.

At a cusp x of F, the face of the rectangular three-box opposite the F-face belongs to a facet F_x. The two rectangles are congruent. If the rectangle of F is nonsquare, so is that of F_x, which cannot be the regular ideal octahedron.

The supporting hyperplanes of F and F_x are asymptotic and meet at exactly the ideal point x. Hence the same F_x cannot occur at a second cusp of F, nor can it meet F across any ridge. Thus all facets in the two lists are distinct and differ from F. Each has defect at least m. Summing their defects in (2) proves the assertion. □

If $P \ncong C$, a positive-defect facet exists: otherwise $W = 0$ in (2), and the volume formula and Kolpakov's unique minimum identify P with C.

4 The two cusp-shape calculations

Lemma 4.1. *The cusp rectangles of the ideal right-angled k-antiprism have aspect ratio $2\cos(\pi/k)$. Thus for $k \geq 4$ all $2k$ cusps are nonsquare.*

Proof. Set $\theta = \pi/k$, $c = \cos\theta$, and take positive square roots in

$$h^2 = \frac{1-c}{1+c}, \quad r^2 = \frac{2c}{1+c}, \quad b^2 = \frac{2}{c(1+c)}, \quad \mu^2 = \frac{2(1-c)}{c}.$$

Use Lorentz form $(-1,1,1,1)$. The two vertex rings have heights $\pm h$, radius r, and angular offset θ. At $u = (1, r\cos\theta, r\sin\theta, h)$, the four incident normals are

$$N = (h,0,0,1), \quad M_0 = (1,b,0,h), \quad M_2 = (1,b\cos 2\theta, b\sin 2\theta, h), \quad L = (1,b\cos\theta, b\sin\theta, -h).$$

Their squares are r^2, μ^2, μ^2, μ^2. Adjacent products vanish; the opposite pairs are parallel. The identities $bc = r$ and $\mu = 2h/r$ give

$$\frac{M_0 + M_2}{\mu} = \frac{2}{\mu}u, \qquad \frac{N}{r} + \frac{L}{\mu} = \left(\frac{h}{r} + \frac{1}{\mu}\right)u.$$

For two opposite unit normals with sum κu, their separation on a fixed horosphere is proportional to κ. To see this, send u to infinity in the upper-half-space model; the mirrors become parallel vertical planes with unit horizontal normals. The two separations therefore have ratio

$$\frac{2/\mu}{h/r + 1/\mu} = \frac{2r^2}{1+h^2} = 2c.$$

The antiprism symmetries are vertex-transitive. Finally $2c > 1$ for $k \geq 4$ (and equals one for $k = 3$). □

Lemma 4.2. *An ideal right-angled three-polyhedron with nine vertices has $d = 3$, $q = 3$ and $s = 3$.*

Proof. The antiprism bound excludes pentagons and larger faces. The complete published nine-vertex octahedrite list consists of one admissible type: two ideal regular octahedra glued across a triangular face [1, Section 4.3, Case D2 and Figure 6]. Rigidity makes this identification geometric. The three gluing cusps have rectangles formed by joining two squares, hence aspect two. The other six cusps remain square. Its three nontriangular faces are quadrilaterals. □

5 Universal exclusion of intermediate volumes

Lemma 5.1. *A positive-defect facet has at least two nontriangular faces. If it has exactly two, it is an antiprism.*

Proof. If its only nontriangle has size k, then $d = k - 3$, whereas the antiprism bound gives $d \geq 2(k-3)$, a contradiction. If the two sizes are $k \geq l \geq 4$, then

$$d = (k-3) + (l-3) \geq 2(k-3).$$

Thus $k = l$ and equality holds in the antiprism bound; its equality statement identifies the facet with A_k. □

Proof of the lower bound in Theorem 1.1. Choose a facet F of smallest positive defect $m \geq 2$. Lemma 3.1 gives $2W \geq m(1+q+s)$. The following cases are exhaustive.

1. If $F = A_k$, then $m = 2k-6$, $q = 2$, and $s = 2k = m+6$ by Lemma 4.1. Hence $2W \geq m(m+9) \geq 22$.
2. A non-antiprism with $m = 2$ is impossible: $v_F = 8$ gives $k \leq 4$ for every face, so its defect is two quadrilaterals and Lemma 5.1 applies.
3. If $m = 3$, Lemma 4.2 gives $2W \geq 3(1+3+3) = 21$.
4. If $m = 4$, every face has size at most five. A pentagon gives equality in the antiprism bound and has already been treated. Otherwise there are four quadrilaterals, so $2W \geq 4(1+4) = 20$.
5. If $m \geq 5$ and F is not an antiprism, Lemma 5.1 gives $q \geq 3$, whence $2W \geq 4m \geq 20$.

Thus $W \geq 10$. Since W is divisible by three, $W \geq 12$. Equation (1) proves the lower bound. In particular the three prospective intermediate defects $3, 6, 9$, with respective volumes $5\pi^2/3$, $2\pi^2$ and $7\pi^2/3$, cannot occur. □

6 Exact attainment by the double

In Lorentz form $(-1,1,1,1,1)$, take the 24 unit normals $(1,a)$, where a runs through the D_4 roots having exactly two nonzero entries, each ± 1. In the time-one Klein section their ideal vertices are the eight $\pm e_i$ and the sixteen $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$. The inequalities are $|x_i| + |x_j| \leq 1$; they are bounded and their origin is strict. Exact enumeration of all $\binom{24}{4} = 10,626$ active four-wall subsets verifies that the displayed vertices are all vertices of the intersection.

Reflect across $n = (1,1,1,0,0)$ by $x \mapsto x - 2\langle x, n\rangle n$. Remove the gluing wall from both cells and identify coincident walls. The resulting 38 halfspaces contain the union. Conversely, on each side of the gluing hyperplane, their inequalities together with that side's gluing inequality include all inequalities of the corresponding original cell. Thus the new intersection equals the union. In particular no unexamined escape directions or additional vertices are being discarded.

The interiors are disjoint, so the volume is twice $4\pi^2/3$. The exact face reconstruction supplies the following independent incidence check.

Quantity	24-cell	Double
Vertices	24	42
Edges	96	168
Ridges	96	164
Facets	24	38
Triangular ridges	96	152
Quadrilateral ridges	0	12
Octahedral facets	24	30
Doubled-octahedron facets	0	8
Ridge defect W	0	12
Reflection-orbifold Euler characteristic	1	2

Each of the eight doubled-octahedron facets has nine vertices, eighteen edges, eight triangles and three quadrilaterals. There are six shared cusps and 36 other cusps. Every cusp has six incident facets, eight edges and twelve ridges. The facet–vertex check is $30 \cdot 6 + 8 \cdot 9 = 6 \cdot 42$. This completes attainment and Theorem 1.1.

7 Verification and formalization scope

The immutable written proof is `results/ideal_ra4_second_volume_20261018.md`. The producer `scripts/fresh_ideal_ra4_double24_volume_20261018.py` uses exact rational arithmetic to reconstruct the original and doubled face lattices, verify Gram conditions at every ridge and cusp, and compute all doubled-facet cusp ratios. The separate checker `scripts/check_ideal_ra4_second_volume_saved_20261018.py` imports no producer; it repeats the construction checks with a separate elimination path and verifies the seven rational-function identities used in the antiprism calculation. Generation, fresh replay, and the final hardened checker all passed. Two independent agent reviewers read the full written proof and both programs and checked the precise published classification premises. They did not independently execute the computation or reconstruct the historical classification census.

Neither program proves the universal geometric injection merely by checking the double. In particular the construction JSON deliberately does not identify its own computation with a proof of the second minimum. The universal quantifiers are supplied by the written proof.

The Lean module `formal/CoxeterProofs/IdealRA4Gap.lean` contains the finite-list budget and integer case-analysis interfaces: given the geometric partition, facet case, and divisibility, the defect gap and volume-numerator gap follow. Its geometric hypotheses are not silently declared published theorems. The Lorentz/cusp injection, shape identification and construction geometry are not yet an end-to-end Lean development. The exact external/internal map and reproduction commands are in `docs/ideal_ra4_second_volume_lemma_map.md` and `docs/ideal_ra4_second_volume_reproduce.md`.

References

[1] A. Kolpakov, *On the optimality of the ideal right-angled 24-cell*, Algebraic & Geometric Topology **12** (2012), 1941–1960. Published 27 October 2012. doi:10.2140/agt.2012.12.1941. Published PDF.